Antenna Characterization

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I. OBJECTIVE

(a) To measure the effect of distance on the power received by the antenna.

(b) To measure the far-field power pattern of the given antenna with respect to angle in both co-polarization and cross-polarization configurations.

(c) To determine the half-power beamwidth and estimate the directivity of the antenna.

(d) To determine the antenna operating frequency and return loss from VSWR measurements.

II. INTRODUCTION

An antenna is a device used for transmitting and receiving electromagnetic waves. In a transmitting system, the antenna converts electrical power into electromagnetic radiation, while in a receiving system it converts the incident electromagnetic wave into an electrical signal. The performance of an antenna is therefore closely related to how its received or radiated power varies with direction, distance, frequency, and polarization.

In this experiment, the characteristics of the given antenna are investigated through measurements of received power as a function of the separation between the transmitting and receiving antennas and as a function of their relative angular orientation. The experiment focuses on the radiation characteristics of the antenna, particularly its directional response and polarization properties. The measurements are carried out using a transmitting antenna connected to an RF signal generator and a receiving antenna connected to a spectrum analyser.

The dependence of received power on the separation between the antennas is first investigated. The distance between the transmitting and receiving antennas is varied while keeping the transmitting frequency, power, antenna height, and polarization fixed. The received power is recorded at different separations and plotted as a function of distance. A second representation is obtained by plotting the received power against $20 \log_{10} R$*, where* R *is the separation between the antennas. This measurement provides an experimental observation of the reduction in received power as the propagation distance is increased.*

The angular response of the antenna is then studied by measuring its beam pattern in two perpendicular planes, namely the E*-plane and the* H*-plane. The* E*-plane is the plane containing the electric-field vector and the direction of propagation, whereas the* H*-plane contains the magnetic-field vector and the direction of propagation. Measurements in these two planes provide two principal cross-sections of the three-dimensional beam pattern of the antenna.*

For the co-polarization measurement, the electric fields of the transmitting and receiving antennas are aligned in the same plane. The receiving antenna is rotated through the specified angular range and the received power is recorded at different angular positions. The resulting measurements are used to obtain the E*-plane and* H*-plane co-polarized beam patterns. The experiment is repeated at additional frequencies so that the change in the beam pattern with frequency can also be examined.*

The cross-polarization response is measured by changing the orientation of the transmitting antenna such that its electric field is perpendicular to that of the receiving antenna. The angular measurements are then repeated under the same experimental conditions used for the co-polarization measurement. A comparison of the two sets of beam patterns allows the response of the antenna to the intended and orthogonal polarization states to be examined. In an ideal case, the response to the orthogonal polarization would be zero; however, a finite response can occur in a practical measurement because of antenna imperfections, alignment errors, and reflections from the surrounding environment.

The co-polarized beam patterns are further used to determine the half-power beamwidth (HPBW) of the antenna. The HPBW is defined as the angular separation between the two points on the main lobe at which the received power falls to half of its maximum value, corresponding to the $-3\,\mathrm{dB}$ *level. The* E*-plane and* H*-plane half-power beamwidths, denoted by* θ_E *and* θ_H*, respectively, provide a measure of the width of the antenna's main beam. A smaller beamwidth corresponds to a more concentrated and directional beam.*

Finally, the measured beamwidths are used to estimate the directivity of the antenna. Directivity describes the concentration of the antenna radiation in a particular direction relative to an isotropic radiator. Using the half-power beamwidth approximation, the directivity is estimated from

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$$D \approx \frac{4\pi}{\theta_E \theta_H}, \tag{1}$$

where θ_E and θ_H are expressed in radians.

Thus, the experiment provides an experimental characterization of the antenna's propagation behaviour, angular beam pattern, polarization response, half-power beamwidth, and directivity. The measurements also allow the beam patterns obtained at different frequencies and for different polarization configurations to be compared and interpreted in terms of the directional properties of the antenna.

III. THEORY

A. Basic Antenna Concepts

An antenna is a transducer that converts electrical signals into electromagnetic waves during transmission and converts incident electromagnetic waves into electrical signals during reception. In the present experiment, one antenna is used as the transmitting antenna and the other as the receiving antenna. The transmitting antenna is connected to an RF signal generator, while the received signal is measured using a spectrum analyser.

The performance of an antenna is described by several characteristics. In this experiment, the main quantities of interest are the variation of received power with distance, the angular radiation pattern, the polarization response, the half-power beamwidth, and the directivity of the antenna.

Two types of antennas relevant to the experiment are the dipole and monopole antennas.

1. Dipole Antenna

A dipole antenna consists of two conducting arms arranged in a straight line and fed at the centre. For a half-wave dipole, the total length of the antenna is approximately

$$L \approx \frac{\lambda}{2}, \tag{2}$$

where λ is the wavelength of the electromagnetic wave. The wavelength is related to the frequency f by

$$\lambda = \frac{c}{f}, \tag{3}$$

where c is the speed of light in free space.

A half-wave dipole does not radiate equally in all directions. Its radiation is strongest in directions perpendicular to the axis of the antenna, while the radiation along the axis of the antenna is comparatively small. This directional behaviour forms the basis for measuring its beam pattern.

2. Monopole Antenna

A monopole antenna is approximately one-half of a dipole mounted above a conducting ground plane. Its length is approximately

$$L \approx \frac{\lambda}{4}. \tag{4}$$

The conducting ground plane acts approximately as a reflecting surface, allowing the monopole to behave similarly to one half of a dipole. As a result, the monopole also exhibits a directional radiation characteristic.

The exact radiation pattern of a practical antenna depends on its physical dimensions, operating frequency, orientation, and surrounding environment.

B. Antenna Radiation Pattern

The radiation pattern describes the variation of the power radiated or received by an antenna as a function of direction. Since the power is not generally the same in every direction, an antenna has a characteristic angular distribution of radiation.

A typical radiation pattern consists of a main lobe together with smaller side and back lobes.

- *The **main lobe** is the region containing the maximum radiation or reception. It indicates the principal direction of the antenna beam.*
- ***Side lobes** are regions of radiation or reception away from the main lobe. Their presence indicates that the antenna has a finite response in directions other than the principal direction.*
- *The **back lobe** is the lobe occurring approximately opposite to the main lobe.*

In a receiving experiment, the radiation pattern can be obtained by measuring the received power while changing the orientation of the receiving antenna. If the transmitting power, frequency, antenna separation, and polarization are kept fixed, the variation in received power with angle provides the angular response of the receiving antenna.

The radiation pattern can be represented using a polar plot. The angular coordinate represents the orientation of the antenna, while the radial coordinate represents the corresponding received power.

C. E-Plane and H-Plane

The electromagnetic field associated with a propagating wave consists of an electric-field vector $\mathbf{E}$ and a magnetic-field vector $\mathbf{H}$, with both fields perpendicular to the direction of propagation.

*The **E-plane** is defined as the plane containing the electric-field vector and the direction of propagation. Similarly, the **H-plane** is the plane containing the magnetic-field vector and the direction of propagation.*

The E-plane and H-plane are perpendicular to one another. Measuring the antenna response in these two planes gives two orthogonal cross-sections of the three-dimensional radiation pattern.

In the experiment, the angular position in the E-plane is represented by θ, while the angular position in the H-plane is represented by ϕ.

The receiving antenna is rotated through the required angular range and the received power is recorded at each angular position. The resulting measurements are then plotted as polar beam patterns.

D. Received Power and Distance

The power received by an antenna depends on the distance between the transmitting and receiving antennas. Let the separation between the two antennas be R and the received power be P_r.

During the distance measurement, the transmitting frequency and amplitude are kept fixed. The transmitting and receiving antennas are also maintained at the same height and with the same polarization. The distance R is then varied and the corresponding received power is recorded.

For ideal free-space propagation, the power density of an electromagnetic wave decreases as the wave spreads over an increasingly large spherical surface. The area of such a surface at a distance R is

$$A = 4\pi R^2. \tag{5}$$

Consequently, the power density decreases approximately as

$$S \propto \frac{1}{R^2}. \tag{6}$$

For fixed transmitting and receiving antenna characteristics, the received power consequently follows the approximate dependence

$$P_r \propto \frac{1}{R^2}. \tag{7}$$

Therefore, increasing the separation between the antennas results in a decrease in received power.

For two different distances R_1 and R_2, the idealized relationship can be written as

$$\frac{P_r(R_2)}{P_r(R_1)} = \left(\frac{R_1}{R_2}\right)^2. \tag{8}$$

In particular, if the distance is doubled,

$$R_2 = 2R_1, \tag{9}$$

then

$$\frac{P_r(2R_1)}{P_r(R_1)} = \frac{1}{4}. \tag{10}$$

The corresponding change in received power, expressed in decibels, is

$$\Delta P_r = 10\log_{10}\left(\frac{1}{4}\right) \approx -6.02 \text{ dB}. \tag{11}$$

Thus, under ideal free-space conditions, doubling the separation produces a decrease of approximately 6 dB in received power.

In the experiment, the measured result may differ from this ideal value due to reflections from nearby objects, imperfect alignment, finite antenna dimensions, and other experimental effects. Therefore, the measured variation is compared with the expected distance dependence rather than assuming that the ideal relationship must hold exactly.

E. Logarithmic Representation of Distance Dependence

The distance dependence can also be examined using a logarithmic distance scale. Taking the logarithm of

$$P_r \propto \frac{1}{R^2} \tag{12}$$

gives

$$\log_{10} P_r = C - 2\log_{10} R, \tag{13}$$

where C represents quantities that remain constant during the measurement.

Multiplying by 10 gives

$$10\log_{10} P_r = C' - 20\log_{10} R. \tag{14}$$

Therefore, a plot involving $20\log_{10} R$ is useful for examining the distance dependence on a logarithmic scale. For an ideal inverse-square dependence, the received power expressed in decibels decreases linearly with $20\log_{10} R$.

This is why, in the experiment, the received power is plotted both directly against R and against $20\log_{10} R$.

F. Far-Field Region

The angular beam pattern of an antenna is measured in the far-field region so that the measured response represents the directional radiation characteristics of the antenna.

In the present experiment, the separation between the transmitting and receiving antennas is maintained at a minimum of

$$R \geq 3\lambda. \tag{15}$$

The antennas are kept at the same height and are aligned appropriately before the angular measurements are performed.

Once the transmitting frequency and power are fixed, the receiving antenna is rotated through different angular positions. The received power at each position is recorded. The resulting angular measurements are used to construct the E-plane and H-plane beam patterns.

G. Polarization of Electromagnetic Waves

Polarization describes the orientation of the electric-field vector of an electromagnetic wave. For a linearly polarized wave, the electric field oscillates predominantly along one fixed direction.

The polarization of the transmitting and receiving antennas plays an important role in determining the received power. Maximum coupling occurs when the receiving antenna is aligned with the polarization of the incident wave.

In the ***co-polarization configuration****, the electric fields of the transmitting and receiving antennas are aligned in the same plane. The receiving antenna is therefore oriented to receive the intended polarization of the transmitted wave.*

In the ***cross-polarization configuration****, the transmitting antenna is adjusted so that its electric field is perpendicular to the electric field of the receiving antenna.*

Ideally, an antenna responding only to one polarization would produce no response for a perfectly orthogonal polarization. In practice, however, a finite cross-polarized response is observed because real antennas are not perfectly polarization-pure. Small alignment errors, reflections from surrounding objects, and imperfections in the antenna can also contribute to the measured signal.

Therefore, comparison of the co-polarized and cross-polarized patterns gives information about the polarization characteristics of the antenna.

H. Co-polarization Beam Pattern

For the co-polarization measurement, the transmitting and receiving antennas are arranged such that their electric fields are parallel. The separation between the antennas is maintained at a minimum of approximately 3λ.

The receiving antenna is first rotated through the E-plane angle θ. The received power is recorded for angular positions ranging from

$$-90^\circ \leq \theta \leq +90^\circ. \tag{16}$$

The antenna is then returned to its reference position and the corresponding H-plane measurement is performed by varying ϕ over the same angular range:

$$-90^\circ \leq \phi \leq +90^\circ. \tag{17}$$

The measured values are plotted as polar patterns. The angular position corresponding to the maximum received power identifies the direction of maximum response.

The co-polarized measurements are repeated for two additional frequencies. This allows the effect of frequency on the shape and width of the beam pattern to be examined.

I. Cross-polarization Beam Pattern

For the cross-polarization measurement, the relative orientation of the antennas is changed so that the electric fields of the transmitting and receiving antennas are perpendicular.

All other relevant measurement conditions are retained as far as possible. In particular, the same frequency, antenna separation, antenna heights, angular intervals, and instrument settings are used as in the co-polarization measurement.

The receiving antenna is again rotated through the E-plane and H-plane angles, and the received power is recorded.

The resulting patterns are compared with the corresponding co-polarized patterns. The difference between the two measurements indicates the extent to which the antenna responds to the intended polarization compared with the orthogonal polarization.

J. Normalized Beam Pattern

For comparing beam patterns, it is useful to express the measured received power relative to the maximum power measured in that particular pattern.

Let $P_r(\theta)$ be the received power at an angular position θ and let $P_{r,\max}$ be the maximum received power. The normalized power is

$$P_{\text{norm}}(\theta) = \frac{P_r(\theta)}{P_{r,\max}}. \tag{18}$$

Expressing this ratio in decibels gives

$$P_{\mathrm{norm,dB}}(\theta) = 10\log_{10}\left(\frac{P_r(\theta)}{P_{r,\max}}\right). \tag{19}$$

At the direction of maximum response,

$$P_r(\theta) = P_{r,\max}, \tag{20}$$

and hence

$$P_{\mathrm{norm,dB}} = 0 \text{ dB}. \tag{21}$$

All other points on the normalized pattern are therefore at values less than or equal to 0 *dB.*

Normalization is particularly useful when comparing measurements obtained at different frequencies or under different polarization configurations because the relative shape of the beam can be compared without being dominated by differences in the absolute received power.

K. Half-Power Beamwidth

The half-power beamwidth (HPBW) is a measure of the angular width of the main lobe of an antenna beam pattern.

Let the maximum received power be $P_{\max}$*. The half-power level is*

$$P_{\mathrm{HP}} = \frac{P_{\max}}{2}. \tag{22}$$

When expressed relative to the maximum power in decibels,

$$P_{\mathrm{HP,dB}} = 10\log_{10}\left(\frac{P_{\max}/2}{P_{\max}}\right). \tag{23}$$

Therefore,

$$P_{\mathrm{HP,dB}} = 10\log_{10}\left(\frac{1}{2}\right) \approx -3.01 \text{ dB}. \tag{24}$$

Hence, the half-power points are conventionally referred to as the -3 *dB points.*

Suppose the two -3 *dB points in the E-plane occur at the angular positions* θ_1 *and* θ_2*. The E-plane half-power beamwidth is*

$$\theta_E = |\theta_2 - \theta_1|. \tag{25}$$

Similarly, if the two -3 *dB points in the H-plane occur at* ϕ_1 *and* ϕ_2*, then the H-plane half-power beamwidth is*

$$\theta_H = |\phi_2 - \phi_1|. \tag{26}$$

The HPBW therefore represents the angular extent of the main lobe between the two directions where the received power has fallen to one-half of its maximum value.

A small HPBW corresponds to a narrow and concentrated beam, whereas a large HPBW corresponds to a broader beam.

In this experiment, the HPBW values are obtained from the co-polarized *E-plane and H-plane beam patterns. The two* -3 *dB points are identified from each pattern and their angular separation is calculated.*

L. Conversion of Beamwidth from Degrees to Radians

The directivity approximation requires the beamwidths to be expressed in radians. The relationship between degrees and radians is

$$180^\circ = \pi \text{ rad}. \tag{27}$$

Therefore, a beamwidth measured in degrees can be converted to radians using

$$\theta_{\mathrm{rad}} = \theta_{\mathrm{deg}}\frac{\pi}{180}. \tag{28}$$

Thus, the measured beamwidths are converted according to

$$\theta_E^{(\mathrm{rad})} = \theta_E^{(\mathrm{deg})}\frac{\pi}{180}, \tag{29}$$

and

$$\theta_H^{(\mathrm{rad})} = \theta_H^{(\mathrm{deg})}\frac{\pi}{180}. \tag{30}$$

These values are then used in the estimation of directivity.

M. Directivity

Directivity is a measure of the concentration of an antenna's radiation in a particular direction compared with an isotropic radiator.

An isotropic radiator is an ideal antenna that radiates equally in all directions. A directional antenna concentrates its radiation into a smaller solid angle. Consequently, a narrower beam generally corresponds to greater directivity.

The directivity of an antenna can be expressed in terms of its beam solid angle Ω_A *as*

$$D = \frac{4\pi}{\Omega_A}. \tag{31}$$

For a practical antenna, the beam solid angle can be approximated using the half-power beamwidths in the two principal planes. The approximation used in this experiment is

$$\Omega_A \approx \theta_E \theta_H, \tag{32}$$

where θ_E and θ_H are expressed in radians.

Substituting this approximation into the expression for directivity gives

$$D \approx \frac{4\pi}{\theta_E \theta_H}. \tag{33}$$

Thus, the measured E-plane and H-plane half-power beamwidths can be used to estimate the directivity of the antenna.

The procedure is therefore to first determine the two -3 dB points from each co-polarized beam pattern, calculate the corresponding HPBW values, convert the beamwidths to radians, and finally substitute them into the above expression.

The relationship also demonstrates the physical significance of beamwidth. If the antenna has a narrow beam in either principal plane, the product $\theta_E \theta_H$ becomes smaller, resulting in a larger value of directivity. Conversely, a broad beam corresponds to a smaller estimated directivity.

N. Effect of Frequency on the Beam Pattern

The wavelength of an electromagnetic wave is related to its frequency by

$$\lambda = \frac{c}{f}. \tag{34}$$

Therefore, changing the operating frequency changes the wavelength while the physical dimensions of the antenna remain unchanged. The ratio of the antenna dimensions to the wavelength consequently changes with frequency.

Since the radiation pattern depends on this ratio, the angular distribution of received power can change when the operating frequency is changed. The main-lobe width, peak response, and side-lobe structure may therefore show differences at different frequencies.

In the experiment, beam patterns are measured at three different frequencies. The measurements are compared to determine how the E-plane and H-plane patterns vary with frequency.

The comparison is meaningful only when other experimental conditions are kept as constant as possible. The antenna separation, antenna height, angular measurement procedure, and instrument settings are therefore retained for the different frequency measurements.

O. Voltage Standing Wave Ratio (VSWR)

When an electromagnetic signal travels through a transmission line and reaches an antenna, a part of the incident signal is radiated while the remaining part may be reflected towards the source. This reflection occurs when the antenna impedance does not match the characteristic impedance of the transmission line.

The incident and reflected waves interfere with each other, producing a standing wave pattern along the transmission line. This pattern consists of alternating voltage maxima and minima.

The Voltage Standing Wave Ratio (VSWR) is defined as

$$VSWR = \frac{V_{\max}}{V_{\min}} \tag{35}$$

where $V_{\max}$ and $V_{\min}$ represent the maximum and minimum voltages, respectively.

For a perfectly matched antenna, there is no reflected wave and the VSWR is equal to 1. A higher VSWR indicates greater impedance mismatch.

1. Reflection Coefficient

The reflection coefficient is represented by Γ. Its magnitude can be calculated from the measured voltages:

$$|\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} \tag{36}$$

The relationship between VSWR and reflection coefficient is

$$VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|} \tag{37}$$

2. Return Loss

Return loss describes the amount of incident power reflected back towards the source. It is expressed in decibels:

$$RL = -20 \log_{10} |\Gamma| \tag{38}$$

A higher return loss indicates better impedance matching.

3. Impedance Bandwidth

The frequency range over which the VSWR remains less than or equal to 2 is commonly considered the acceptable impedance bandwidth.

The bandwidth is

$$BW = f_H - f_L \tag{39}$$

The fractional bandwidth is

$$BW_{\text{fractional}} = \frac{f_H - f_L}{f_0} \times 100\% \tag{40}$$

where the centre frequency is

$$f_0 = \frac{f_H + f_L}{2} \tag{41}$$

Thus, VSWR and return loss measurements provide information about antenna impedance matching and operating bandwidth.

P. Summary of Relevant Relations

The main theoretical relations used in Parts B–E of the experiment are summarized below:

$$\lambda = \frac{c}{f}, \tag{42}$$

$$P_r \propto \frac{1}{R^2}, \tag{43}$$

$$\frac{P_r(R_2)}{P_r(R_1)} = \left(\frac{R_1}{R_2}\right)^2, \tag{44}$$

$$\Delta P_r = 10 \log_{10}\left(\frac{P_r(R_2)}{P_r(R_1)}\right), \tag{45}$$

$$P_{\text{norm}} = \frac{P_r}{P_{r,\max}}, \tag{46}$$

$$P_{\text{norm,dB}} = 10 \log_{10}\left(\frac{P_r}{P_{r,\max}}\right), \tag{47}$$

$$\theta_E = |\theta_2 - \theta_1|, \tag{48}$$

$$\theta_H = |\phi_2 - \phi_1|, \tag{49}$$

$$\theta_{\text{rad}} = \theta_{\text{deg}} \frac{\pi}{180}, \tag{50}$$

$$D \approx \frac{4\pi}{\theta_E \theta_H}. \tag{51}$$

These relations provide the theoretical basis for analysing the measurements performed in the experiment. The distance measurements are used to study the variation of received power with antenna separation, while the angular measurements provide the co-polarized and cross-polarized beam patterns. The co-polarized patterns are subsequently used to determine the half-power beamwidths and estimate the directivity of the antenna.

IV. COMPONENTS REQUIRED

The following components and instruments were used for the antenna characterization experiment:

- *Monopole and dipole antennas*
- *RF signal generator*
- *Digital storage oscilloscope (DSO)*
- *Spectrum analyser*
- *RF Tee connector*
- *50 Ω termination*
- *Antenna support stand*
- *Coaxial cables and suitable RF adapters*

V. EXPERIMENTAL SETUP

The experimental setup used for the antenna characterization measurements consists of a transmitting antenna, a receiving antenna, an RF signal generator, a spectrum analyser, antenna support stands, and the required coaxial cables and RF adapters. The transmitting antenna is connected to the RF signal generator, while the receiving antenna is connected to the spectrum analyser. Both antennas are mounted on support stands so that they remain at the same height during the measurements.

The RF signal generator is used to provide a continuous signal at the selected frequency. This signal is transmitted by the transmitting antenna and received by the second antenna. The received power is measured using the spectrum analyser. The separation between the transmitting and receiving antennas is denoted by R and is varied during the distance-dependence measurements.

For the beam-pattern measurements, the separation between the two antennas is kept sufficiently large for the far-field condition specified in the experiment. The antennas are first arranged with their electric fields in the same plane for the co-polarization measurements. The receiving antenna is then rotated through the required angular range while the received power is recorded. Measurements are made in both the E-plane and the H-plane to obtain the corresponding beam patterns.

For the cross-polarization measurements, the orientation of the transmitting antenna is changed so that its electric field is perpendicular to that of the receiving antenna. The same angular measurements are then repeated while keeping the separation, frequency, antenna heights, and instrument settings unchanged. The co-polarized and cross-polarized measurements can therefore be compared directly.

The basic arrangement of the transmitting and receiving antennas is shown in Fig. 1.

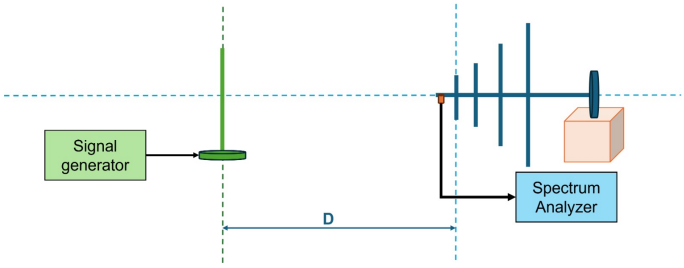


FIG. 1: Experimental setup for antenna characterization.

During the experiment, care was taken to maintain the same antenna height and to keep the required alignment while changing the distance or angular orientation. The signal generator frequency and amplitude were kept fixed for each set of measurements, and the received power was recorded from the spectrum analyser.

A. VSWR Measurement Setup

A separate experimental arrangement was used to measure the Voltage Standing Wave Ratio (VSWR) and return loss of the antenna. The setup consisted of an RF signal generator, an RF Tee connector, the antenna under test (DUT), a digital storage oscilloscope, and coaxial cables.

The output of the RF signal generator was connected to the RF Tee connector. The oscilloscope was connected to the measurement branch of the Tee, while the antenna under test was connected to the other branch.

The oscilloscope input impedance was set to 1 Megaohm. The signal generator was operated at a constant output amplitude during the measurements. When the incident signal reached the antenna, a portion of the signal was reflected due to impedance mismatch. The incident and reflected waves produced voltage variations, which were observed using the oscilloscope.

At each selected frequency, a small frequency sweep was performed to observe the voltage variations. The maximum and minimum voltage values were recorded from the oscilloscope. These measurements were subsequently used to calculate VSWR, reflection coefficient, and return loss.

The measurements were repeated at different frequencies to investigate the impedance-matching characteristics of the antenna.

VI. OBSERVATIONS

The measurements were carried out by varying the separation between the transmitting and receiving antennas and by changing the angular orientation of the antenna. The input power of the signal generator was maintained at 0 dBm *during the measurements. The received output power was recorded using the spectrum analyser.*

A. Distance Dependence of Received Power

For the distance-dependence measurement, the separation between the two antennas was varied while keeping the input power fixed at 0 dBm. *The measured received power for the three different separations is given in Table I.*

TABLE I: Variation of received power with antenna separation.

Distance, R (m)	P_{in} (dBm)	P_{out} (dBm)
8.12	0	-61.2
7.35	0	-54.7
6.58	0	-51.2

It can be observed that the received power increases as the separation between the antennas is reduced. The received power changes from -61.2 dBm *at a separation of* 8.12 m *to* -51.2 dBm *at* 6.58 m. *Thus, the measurement shows the expected decrease in received signal strength with increasing propagation distance.*

B. Co-polarization Measurement

The co-polarization measurements were performed at a fixed frequency of 1.3 GHz *with an input power of* 0 dBm. *The receiving antenna was rotated through the specified angular positions and the corresponding received power was recorded.*

1. Co-polarization: H-plane

The measured received power as a function of angle in the H-plane is given in Table II.

TABLE II: Co-polarization measurement in the H-plane at 1.3 GHz.

θ (deg)	P_{in} (dBm)	P_{out} (dBm)
0	0	-51.8
20	0	-60.5
40	0	-66.9
60	0	-75.0
80	0	-75.0
100	0	-70.0
120	0	-73.9
140	0	-66.5
160	0	-65.2
180	0	-67.13
200	0	-64.5
220	0	-70.5
240	0	-74.7
260	0	-63.6
280	0	-61.5
300	0	-60.8
320	0	-68.3
340	0	-62.3
360	0	-58.8

The maximum recorded received power in this measurement is $-51.8\,\text{dBm}$ *at* $0°$*. The received power generally decreases as the antenna is rotated away from this orientation, reaching* $-75.0\,\text{dBm}$ *at both* $60°$ *and* $80°$*. The power then varies with angular position and increases again towards* $360°$*, where the recorded value is* $-58.8\,\text{dBm}$*. The variation of received power with angle represents the directional response of the antenna in the H-plane.*

2. Co-polarization: E-plane

The corresponding co-polarization measurements in the E-plane are shown in Table III.

TABLE III: Co-polarization measurement in the E-plane at 1.3 GHz.

θ (deg)	P_{in} (dBm)	P_{out} (dBm)
0	0	-53.0
20	0	-54.1
40	0	-55.3
60	0	-56.4
80	0	-61.2
100	0	-64.1
120	0	-61.02
140	0	-70.4
160	0	-68.8
180	0	-61.7
200	0	-60.1
220	0	-70.3
240	0	-58.1
260	0	-55.1
280	0	-56.0
300	0	-57.6
320	0	-61.4
340	0	-57.4
360	0	-53.4

For the E-plane measurement, the highest received power is $-53.0\,\text{dBm}$ *at* $0°$*. The received power initially decreases with increasing angle, reaching* $-56.4\,\text{dBm}$ *at* $60°$ *and* $-70.4\,\text{dBm}$ *at* $140°$*. The power then varies with angle and shows several fluctuations before increasing again towards* $360°$*, where the recorded value is* $-53.4\,\text{dBm}$*. The lowest recorded received power is* $-70.4\,\text{dBm}$ *at* $140°$*. The overall angular variation represents the directional response of the antenna in the E-plane.*

C. Cross-polarization Measurement

For the cross-polarization measurements, the transmitting and receiving antennas were arranged with their polarization directions perpendicular to each other. The measurements were performed at the same frequency of 1.3 GHz *and with an input power of* $0\,\text{dBm}$*.*

1. Cross-polarization: H-plane

The measured cross-polarized received power in the H-plane is given in Table IV.

TABLE IV: Cross-polarization measurement in the H-plane at 1.3 GHz.

θ (deg)	P_{in} (dBm)	P_{out} (dBm)
0	0	-65.5
20	0	-76.5
40	0	-81.5
60	0	-70.0
80	0	-73.9
100	0	-71.0
120	0	-65.0
140	0	-73.2
160	0	-71.5
180	0	-70.25
200	0	-69.8
220	0	-69.2
240	0	-72.9
260	0	-64.2
280	0	-63.7
300	0	-61.1
320	0	-59.2
340	0	-60.2
360	0	-63.7

The received power varies considerably with angular position. The lowest recorded power is $-81.5\,\text{dBm}$ *at* 40°*, while the highest recorded value is* $-59.2\,\text{dBm}$ *at* 320°*. A finite received signal is observed throughout the angular range even though the measurement was performed in the cross-polarized configuration.*

2. Cross-polarization: E-plane

The cross-polarized E-plane measurements are given in Table V.

TABLE VI: Measured VSWR and return loss at different test frequencies.

Frequency	V_{max}	V_{min}	VSWR	$\lvert\Gamma\rvert$	RL (dB)
120 MHz	4.50 V	0.40 V	11.25	0.837	1.55
300 MHz	3.00 V	2.00 V	1.50	0.200	13.98

TABLE V: Cross-polarization measurement in the E-plane at 1.3 GHz.

θ (deg)	P_{in} (dBm)	P_{out} (dBm)
0	0	-64.71
20	0	-63.79
40	0	-64.67
60	0	-65.10
80	0	-71.34
100	0	-73.50
120	0	-74.68
140	0	-80.47
160	0	-75.21
180	0	-77.48
200	0	-81.37
220	0	-77.46
240	0	-75.11
260	0	-70.75
280	0	-69.52
300	0	-68.67
320	0	-64.21
340	0	-63.05
360	0	-64.16

In the cross-polarized E-plane measurement, the highest recorded received power is $-63.05\,\text{dBm}$ *at* 340°*, while the lowest recorded value is* $-81.37\,\text{dBm}$ *at* 200°*. The received power remains relatively low over a large part of the angular range, with a gradual increase towards the* 320°–360° *region. Some fluctuations are also present in the measured values. The finite received power observed in this configuration indicates that some cross-polarized signal was detected during the measurement.*

D. VSWR and Return Loss Observations

The VSWR measurements were carried out using the RF signal generator, RF Tee connector, antenna under test, and digital storage oscilloscope. The oscilloscope was used to observe the voltage variations produced by the incident and reflected signals.

At each selected frequency, a small frequency sweep was performed and the maximum and minimum voltage values were recorded. These values were used to calculate the Voltage Standing Wave Ratio (VSWR), reflection coefficient, and return loss.

The measurements obtained at the two selected frequencies are summarized in Table VI.

At 120 MHz, the measured VSWR was 11.25, indicating considerable impedance mismatch. The corresponding reflection coefficient was 0.837, with a return loss of approximately 1.55 dB.

At 300 MHz, the measured VSWR was 1.50, which indicates considerably better impedance matching compared to the measurement at 120 MHz. The reflection coefficient decreased to 0.200, while the return loss increased to approximately 13.98 dB.

The comparison shows that the antenna exhibits different impedance-matching characteristics at different frequencies. Among the two measured frequencies, the antenna showed better matching at 300 MHz, as indicated by its lower VSWR and higher return loss.

Since measurements were available only at these two frequencies, the exact resonant frequency and the complete impedance bandwidth could not be determined from these observations alone.

E. Overall Observation

The measurements show that the received power depends on both the separation between the antennas and their relative angular orientation. For the distance measurement, an increase in separation results in a decrease in the received power. The co-polarization measurements show clear angular variations in both the E- and H-planes, demonstrating the directional response of the antenna.

The cross-polarization measurements also show a finite received signal over the measured angular range. In the cross-polarized E-plane, the received power is generally lower than the corresponding co-polarized response, although the measured values vary with angle. Ideally, for perfectly orthogonal polarization, the received power would be very small. The finite values observed in the experiment can be associated with practical effects such as imperfect polarization alignment, antenna imperfections, reflections, and other experimental limitations.

The measured angular data can be used to obtain the polar radiation patterns for both co- and cross-polarization. The co-polarized E- and H-plane patterns can further be used to estimate the half-power beamwidth and directivity of the antenna.

VII. GRAPHS

The measured data were plotted to study the variation of received power with the separation between the transmitting and receiving antennas and with the angular orientation of the receiving antenna. The angular measurements were normalized with respect to the maximum measured power in each dataset. Thus, the maximum value of each beam pattern is represented by 0 dB.

A. Received Power as a Function of Distance

The variation of received power with antenna separation is shown in Fig. 2. The received power decreases as the separation between the transmitting and receiving antennas increases. In the measured range, the received power changes from about $-51.2\,\mathrm{dBm}$ *at* $R = 6.58\,\mathrm{m}$ *to* $-61.2\,\mathrm{dBm}$ *at* $R = 8.12\,\mathrm{m}$.

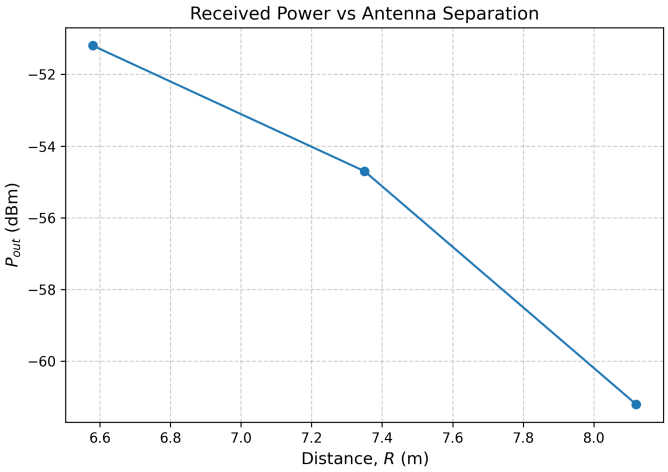


FIG. 2: Variation of received power with antenna separation.

B. Received Power as a Function of $20\log_{10}(R)$

The received power was also plotted against $20\log_{10}(R)$ *to examine the distance dependence on a logarithmic scale. The resulting plot is shown in Fig. 3. The measured points show a decreasing trend in received power with increasing* $20\log_{10}(R)$*, which is consistent with the reduction in received signal strength as the propagation distance increases.*

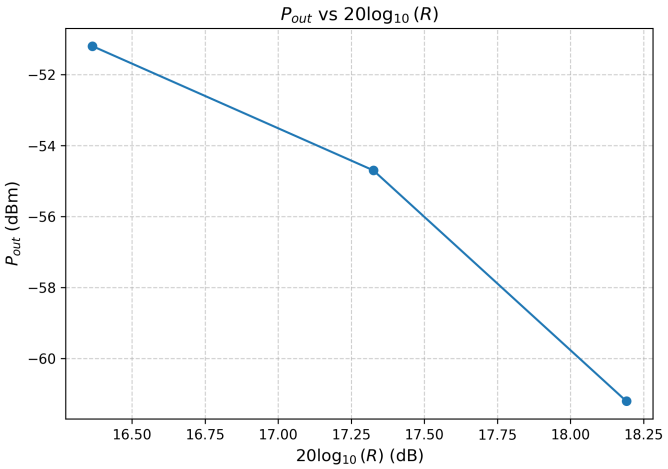


FIG. 3: Received power plotted against $20\log_{10}(R)$.

C. Co-polarization Beam Patterns

The co-polarization measurements were carried out at a frequency of 1.3 GHz *with the transmitting and receiving antennas having the same polarization orientation. The resulting H-plane and E-plane beam patterns are shown in Figs. 4 and 5, respectively.*

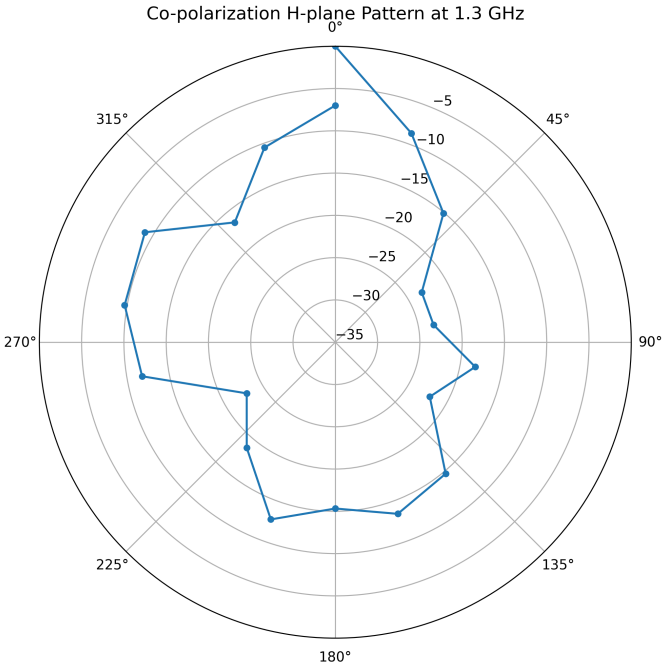


FIG. 4: Normalized co-polarization beam pattern in the H-plane at 1.3 GHz.

The co-polarization H-plane pattern shows its maximum response near 0°*, followed by a decrease in received power as the antenna is rotated away from this direction. The pattern also shows several variations at other angles, indicating that the measured response is not perfectly symmetric.*

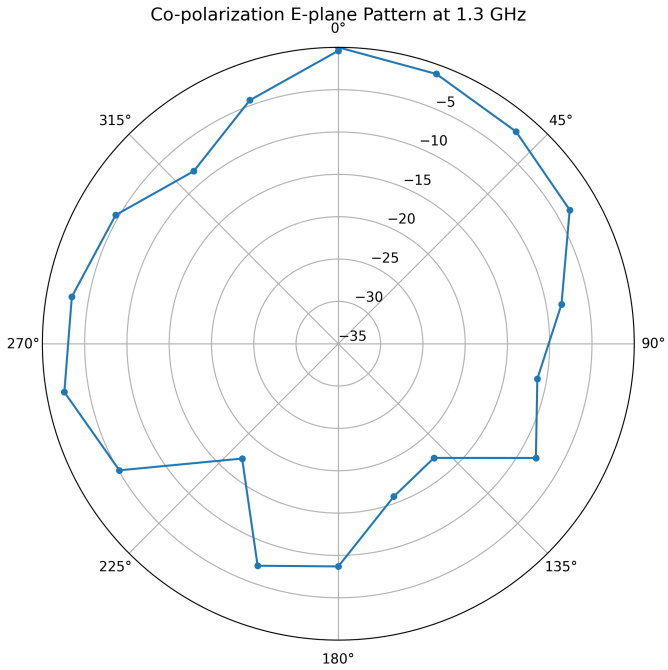


FIG. 5: Normalized co-polarization beam pattern in the E-plane at 1.3 GHz.

The E-plane pattern has a broader main response around the 0° *direction. The received power decreases towards the lower part of the pattern and then increases again towards larger angles. The measured E-plane pattern is therefore considerably broader than the H-plane pattern.*

D. Cross-polarization Beam Patterns

For the cross-polarization measurements, the polarization directions of the transmitting and receiving antennas were made perpendicular to each other. The resulting H-plane and E-plane patterns at 1.3 GHz *are shown in Figs. 6 and 7.*

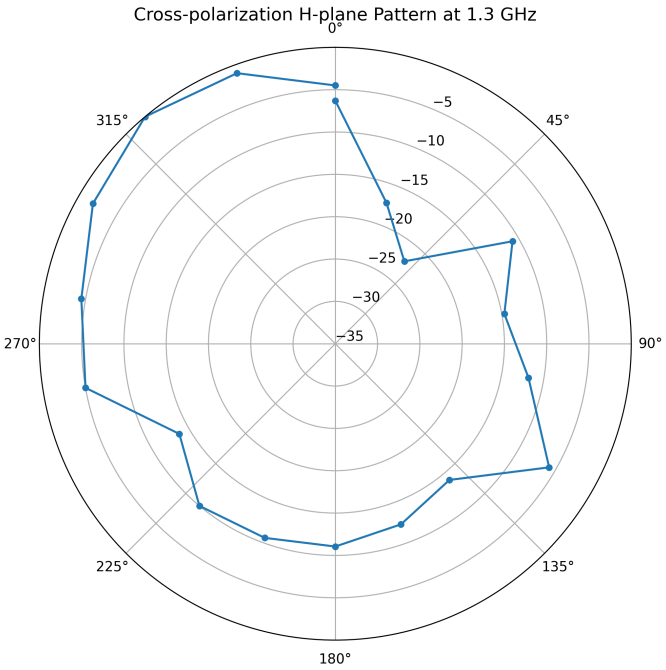


FIG. 6: Normalized cross-polarization beam pattern in the H-plane at 1.3 GHz.

The cross-polarization H-plane pattern has a measurable response over the full angular range. Its maximum occurs away from the 0° direction, and the pattern contains noticeable angular variations. This shows that the cross-polarized response is not completely suppressed in the experimental setup.

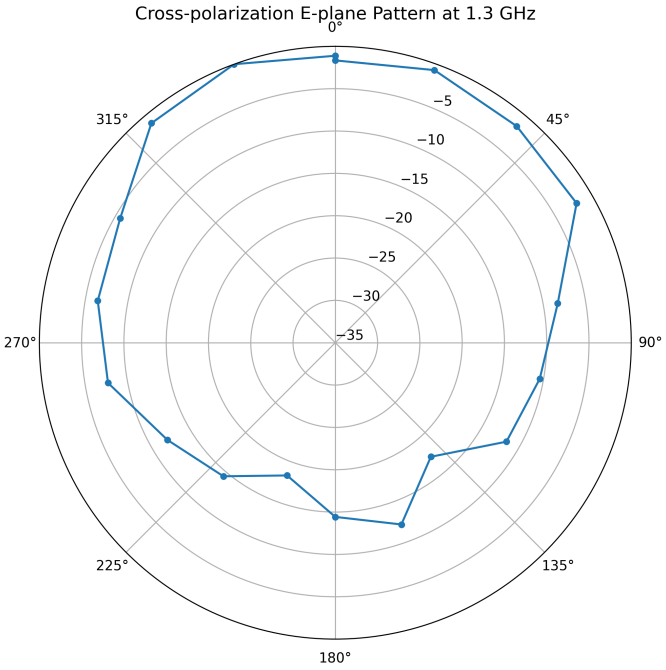


FIG. 7: Normalized cross-polarization beam pattern in the E-plane at 1.3 GHz.

The cross-polarization E-plane pattern shows a relatively strong response over a broad angular region. The response decreases around the lower part of the pattern and increases again towards the opposite side. The presence of this finite response indicates that the polarization isolation in the practical setup is not ideal.

E. Comparison of Co- and Cross-polarization Patterns

The co-polarization and cross-polarization measurements were plotted together for both planes to compare their angular behaviour. The H-plane comparison is shown in Fig. 8.

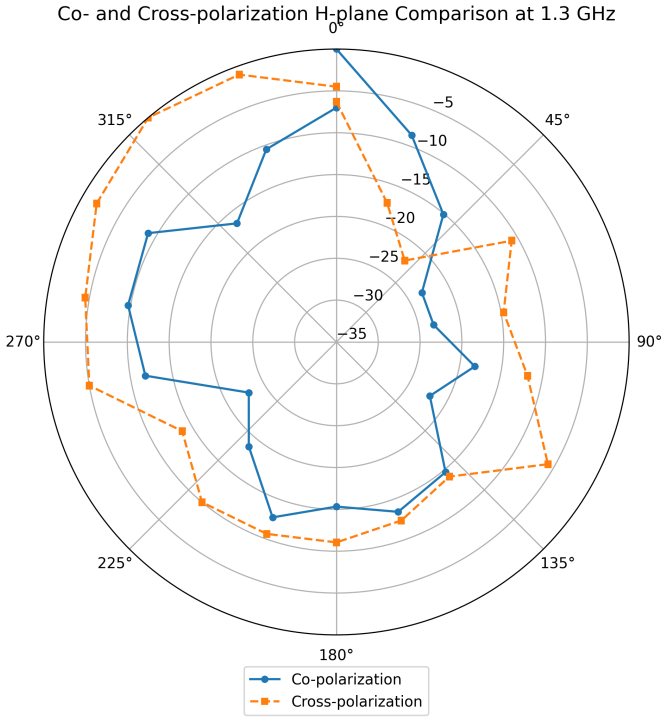


FIG. 8: Comparison of normalized co- and cross-polarization patterns in the H-plane at 1.3 GHz.

In the H-plane comparison, the co- and cross-polarization patterns have different angular shapes. The co-polarization response has its maximum near 0°, whereas the cross-polarization response is relatively stronger at other angular positions. The two patterns therefore show a clear difference in their angular response.

The corresponding comparison for the E-plane is shown in Fig. 9.

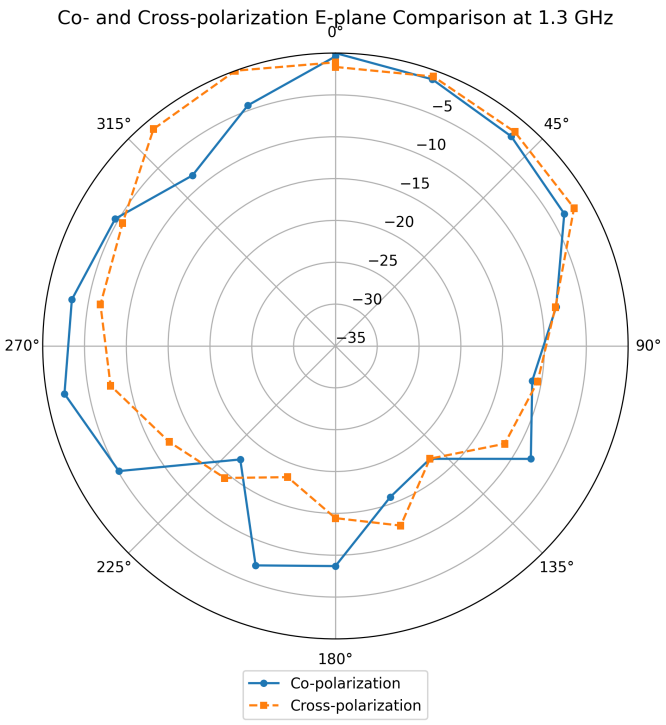


FIG. 9: Comparison of normalized co- and cross-polarization patterns in the E-plane at 1.3 GHz.

The E-plane comparison shows that the co- and cross-polarization patterns have broadly similar angular trends over some regions, although noticeable differences are present at several angles. Since each pattern has been normalized separately, this comparison is used mainly to examine the shape and angular variation of the responses rather than their absolute power levels.

The co-polarization patterns are used for determining the half-power beamwidth of the antenna. The $-3\,\mathrm{dB}$ *points obtained from the measured E- and H-plane patterns are used to determine the corresponding beamwidths and to estimate the antenna directivity.*

F. VSWR

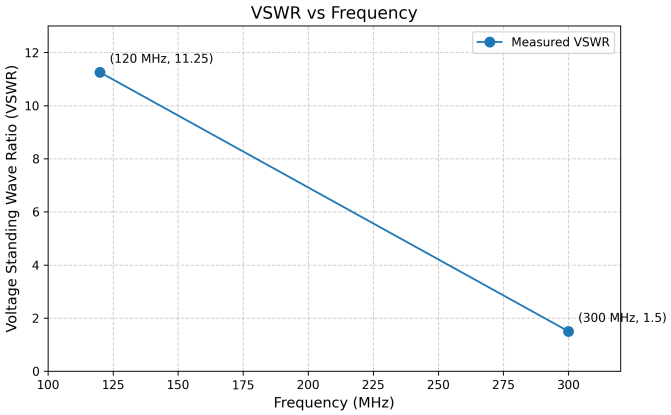


FIG. 10: Variation of voltage standing wave ratio (VSWR) with frequency based on the measured values at 120 MHz and 300 MHz.

The VSWR versus frequency plot shows a decrease in VSWR from 11.25 at 120 MHz to 1.50 at 300 MHz. At 120 MHz, the high VSWR indicates a considerable impedance mismatch between the antenna and the transmission line, resulting in a larger reflected signal. In comparison, the lower VSWR at 300 MHz indicates better impedance matching and a smaller reflection coefficient. The corresponding return loss also increases from 1.55 dB at 120 MHz to 13.98 dB at 300 MHz, supporting this observation.

Thus, among the two measured frequencies, better impedance matching was obtained at 300 MHz. However, since measurements were taken at only two frequencies, the plot is insufficient to identify the exact resonant frequency or determine the impedance bandwidth of the antenna.

VIII. CALCULATIONS AND ERROR ANALYSIS

A. Distance Dependence

For the distance measurements, the received power was recorded at three different antenna separations. The quantity $20\log_{10}(R)$ *was calculated for each distance using*

$$20\log_{10}(R). \tag{52}$$

For example, for $R = 8.12\,\mathrm{m}$,

$$20\log_{10}(8.12) = 18.19\,\mathrm{dB}. \tag{53}$$

Similarly,

$$20\log_{10}(7.35) = 17.33\,\text{dB}, \tag{54}$$
$$20\log_{10}(6.58) = 16.36\,\text{dB}. \tag{55}$$

The measured values were therefore

TABLE VII: Distance measurements and calculated logarithmic distance.

R (m)	$20\log_{10}(R)$ (dB)	P_{out} (dBm)
6.58	16.36	-51.2
7.35	17.33	-54.7
8.12	18.19	-61.2

The received power decreases as the antenna separation increases. A linear fit of P_{out} *against* $20\log_{10}(R)$ *gives approximately*

$$P_{\text{out}} \approx -5.44\,(20\log_{10} R) + 38.39, \tag{56}$$

with a coefficient of determination of approximately

$$R^2 \approx 0.96. \tag{57}$$

Thus, the measurements show a reasonably clear dependence of received power on distance. However, only three distance points were measured, so the fitted slope should be regarded only as an indication of the observed trend rather than a precise experimental determination of the propagation law.

For comparison, an ideal free-space inverse-square dependence would give a power reduction of

$$20\log_{10}(2) = 6.02\,\text{dB} \tag{58}$$

when the distance is doubled. The measured distances do not contain an exact doubling pair, so this reduction cannot be directly verified from the present data.

B. Half-Power Beamwidth

The half-power beamwidth (HPBW) was determined from the co-polarization patterns using the $-3\,\text{dB}$ *level relative to the maximum received power. Since the measurements are given in dBm, a reduction of* $3\,\text{dB}$ *from the maximum measured power was used to locate the half-power points.*

1. E-plane HPBW

For the co-polarization E-plane pattern, the maximum measured power is

$$P_{\max} = -63.05\,\text{dBm}. \tag{59}$$

Therefore, the $-3\,\text{dB}$ *level is*

$$P_{-3\,\text{dB}} = -63.05 - 3 = -66.05\,\text{dBm}. \tag{60}$$

The first $-3\,\text{dB}$ *crossing lies between* 60° *and* 80°*, where the measured powers are* $-65.10\,\text{dBm}$ *and* $-71.34\,\text{dBm}$*, respectively. Linear interpolation gives*

$$\theta_1 = 60 + \frac{-66.05 - (-65.10)}{-71.34 - (-65.10)}(20), \tag{61}$$

which gives

$$\theta_1 \approx 63.04^\circ. \tag{62}$$

On the other side of the main lobe, the $-3\,\text{dB}$ *crossing lies between* 300° *and* 320°*. Using the corresponding measured powers of* $-68.67\,\text{dBm}$ *and* $-64.21\,\text{dBm}$*,*

$$\theta_2 = 300 + \frac{-66.05 - (-68.67)}{-64.21 - (-68.67)}(20), \tag{63}$$

giving

$$\theta_2 \approx 311.75^\circ. \tag{64}$$

Since the main lobe crosses the $0^\circ/360^\circ$ *boundary, the E-plane HPBW is*

$$\theta_E = (360^\circ - \theta_2) + \theta_1. \tag{65}$$

Hence,

$$\boxed{\theta_E \approx 111.3^\circ}. \tag{66}$$

For use in the directivity calculation, the angle is converted into radians:

$$\theta_E = 111.3^\circ \left(\frac{\pi}{180^\circ}\right) \approx 1.942\,\text{rad}. \tag{67}$$

2. H-plane HPBW

For the co-polarization H-plane pattern, the maximum measured power is

$$P_{\max} = -51.8\,\text{dBm}. \tag{68}$$

The corresponding $-3\,\text{dB}$ *level is therefore*

$$P_{-3\,\mathrm{dB}} = -51.8 - 3 = -54.8\,\mathrm{dBm}. \tag{69}$$

The measured power at 0° *is* $-51.8\,\mathrm{dBm}$*, while at* 20° *it is* $-60.5\,\mathrm{dBm}$*. Therefore, the first crossing of the* $-3\,\mathrm{dB}$ *level occurs between these two angles. Linear interpolation gives*

$$\theta_1 = 0 + \frac{-54.8 - (-51.8)}{-60.5 - (-51.8)}(20), \tag{70}$$

so that

$$\boxed{\theta_1 \approx 6.90^\circ}. \tag{71}$$

A second clear $-3\,\mathrm{dB}$ *crossing could not be obtained from the measured H-plane data. In particular, the measurements near* 340°–360° *do not return to the same level as the* 0° *measurement. The values at* 340° *and* 360° *are approximately* $-62.3\,\mathrm{dBm}$ *and* $-58.8\,\mathrm{dBm}$*, respectively, compared with* $-51.8\,\mathrm{dBm}$ *at* 0°*.*

Therefore, the H-plane HPBW cannot be determined reliably from two directly measured $-3\,\mathrm{dB}$ *crossings. If approximate symmetry of the main lobe about* 0° *is assumed, the HPBW can be estimated as*

$$\theta_H \approx 2(6.90^\circ) = 13.8^\circ. \tag{72}$$

Thus,

$$\boxed{\theta_H \approx 13.8^\circ} \tag{73}$$

is used only as an approximate value in the directivity calculation.

In radians,

$$\theta_H = 13.8^\circ \left(\frac{\pi}{180^\circ}\right) \approx 0.241\,\mathrm{rad}. \tag{74}$$

C. Estimation of Directivity

The approximate directivity of the antenna can be estimated from the E- and H-plane beamwidths using

$$D \approx \frac{4\pi}{\theta_E \theta_H}, \tag{75}$$

where the beamwidths must be expressed in radians. Using

$$\theta_E = 1.942\,\mathrm{rad} \tag{76}$$

and

$$\theta_H \approx 0.241\,\mathrm{rad}, \tag{77}$$

we obtain

$$D \approx \frac{4\pi}{(1.942)(0.241)} \tag{78}$$

$$\approx 26.9. \tag{79}$$

Therefore, the estimated directivity is

$$\boxed{D \approx 26.9}. \tag{80}$$

Expressed in decibels,

$$D_{\mathrm{dBi}} = 10\log_{10}(26.9) \approx 14.3\,\mathrm{dBi}. \tag{81}$$

Hence,

$$\boxed{D \approx 14.3\,\mathrm{dBi}}. \tag{82}$$

Since the H-plane HPBW was obtained using an assumed symmetry rather than two clearly measured $-3\,\mathrm{dB}$ *crossings, the calculated directivity should be considered an approximate estimate.*

D. VSWR

The VSWR, reflection coefficient and return loss were calculated using the measured maximum and minimum voltages.

The expressions used are:

$$VSWR = \frac{V_{\max}}{V_{\min}}$$

$$|\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$$

$$RL = -20\log_{10}|\Gamma| \ dB$$

E. At 120 MHz

Measured values:
Vmax = 4.50 V Vmin = 0.40 V
Calculation of VSWR:

$$VSWR = \frac{4.50}{0.40} = 11.25$$

Calculation of reflection coefficient:

$$|\Gamma| = \frac{4.50 - 0.40}{4.50 + 0.40}$$

$$|\Gamma| = \frac{4.10}{4.90} \approx 0.837$$

Calculation of return loss:

$$RL = -20\log_{10}(0.837)$$

$$RL \approx 1.55\ dB$$

F. At 300 MHz

Measured values:
Vmax = 3.00 V Vmin = 2.00 V
Calculation of VSWR:

$$VSWR = \frac{3.00}{2.00} = 1.50$$

Calculation of reflection coefficient:

$$|\Gamma| = \frac{3.00 - 2.00}{3.00 + 2.00}$$

$$|\Gamma| = \frac{1.00}{5.00} = 0.200$$

Calculation of return loss:

$$RL = -20\log_{10}(0.200)$$

$$RL \approx 13.98\ dB$$

G. Summary of Calculated Values

TABLE VIII: Calculated VSWR, reflection coefficient and return loss

Frequency (MHz)	VSWR	$\|\Gamma\|$	Return Loss (dB)
120	11.25	0.837	1.55
300	1.50	0.200	13.98

The results indicate a considerable impedance mismatch at 120 MHz, as reflected by the high VSWR and low return loss. At 300 MHz, the VSWR is lower and the return loss is higher, indicating better impedance matching.

H. Sources of Error and Experimental Uncertainty

The main uncertainty in the beamwidth calculation arises from the angular sampling interval of 20°*. The exact positions of the* $-3\,\mathrm{dB}$ *crossings therefore cannot be known more accurately than the spacing of the measured points unless a finer angular scan is performed. Linear interpolation between adjacent points gives a better estimate of the crossing, but it does not remove the uncertainty associated with the discrete measurements.*

A particularly important limitation is seen in the H-plane data. The measurements at 0° *and* 360° *do not agree, even though they represent the same physical orientation. The difference is approximately*

$$|-51.8 - (-58.8)| = 7.0\,\mathrm{dB}. \tag{83}$$

This discrepancy may be caused by small changes in antenna alignment, mechanical positioning, cable movement, reflections from nearby objects, or fluctuations in the received signal. Because of this, the second $-3\,\mathrm{dB}$ *point in the H-plane could not be identified confidently.*

The distance measurement also has some limitations. Only three antenna separations were used, and the antennas may not have been perfectly aligned at every position. Reflections from the surroundings can also produce multipath interference and affect the measured received power. These effects can explain why the measured variation does not exactly follow an ideal free-space dependence.

The directivity estimate is especially sensitive to the uncertainty in the beamwidths because it depends on the product $\theta_E\theta_H$*. Therefore, any error in the H-plane beamwidth directly affects the calculated directivity. The value* $D \approx 26.9$*, or* $14.3\,\mathrm{dBi}$*, should consequently be treated as an estimate based on the symmetry assumption for the H-plane pattern rather than as a precise measured directivity.*

IX. RESULTS

The antenna characterization measurements were performed at a frequency of $1.3\,\mathrm{GHz}$*. The measured distance dependence and the main beamwidth and directivity results are summarized below.*

TABLE IX: Measured received power for different antenna separations.

R (m)	$20\log_{10}(R)$ (dB)	P_{out} (dBm)
6.58	16.36	-51.2
7.35	17.33	-54.7
8.12	18.19	-61.2

The received power decreases as the separation between the transmitting and receiving antennas increases. The

three measurements therefore show the expected decrease in received power with increasing distance.

TABLE X: Summary of antenna characterization results, including beamwidth, directivity and VSWR measurements.

Quantity	Result
Beamwidth and Directivity at 1.3 GHz	
E-plane HPBW, θ_E	111.3°
E-plane HPBW	1.942 rad
H-plane -3 dB crossing	6.90° from 0°
H-plane HPBW	Not reliably determined
Approx. H-plane HPBW	13.8°
Estimated directivity, D	26.9
Estimated directivity	14.3 dBi
VSWR Measurements	
VSWR at 120 MHz	11.25
Reflection coefficient at 120 MHz	0.837
Return loss at 120 MHz	1.55 dB
VSWR at 300 MHz	1.50
Reflection coefficient at 300 MHz	0.200
Return loss at 300 MHz	13.98 dB

The E-plane pattern gives a half-power beamwidth of 111.3°*, or* 1.942 rad*. In the H-plane, one* -3 dB *crossing was obtained at approximately* 6.90° *from the* 0° *reference direction. A second clear crossing was not obtained from the measured data, so the H-plane HPBW could not be determined directly. Assuming approximate symmetry of the main lobe gives an H-plane HPBW of about* 13.8°*.*

Using this approximate H-plane beamwidth with the measured E-plane beamwidth, the antenna directivity was estimated as $D \approx 26.9$*, or* 14.3 dBi*. Since the H-plane beamwidth is based on a symmetry assumption, the directivity should also be regarded as an approximate result.*

The VSWR measurements showed different impedance-matching conditions at the two tested frequencies. At 120 MHz*, the measured VSWR was* 11.25*, with a reflection coefficient of* 0.837 *and a return loss of* 1.55 dB*, indicating poor impedance matching. At* 300 MHz*, the VSWR was considerably lower at* 1.50*, while the reflection coefficient was* 0.200 *and the return loss was* 13.98 dB*. This indicates better impedance matching at* 300 MHz *compared with* 120 MHz*.*

Overall, the measurements show that the antenna has a broad E-plane radiation pattern, while the estimated H-plane beamwidth suggests a narrower main lobe in that plane. The VSWR results also demonstrate that the impedance matching varies with frequency. However, the two VSWR measurements alone are not sufficient to determine the exact resonant frequency or the complete impedance bandwidth of the antenna.

X. CONCLUSION

The antenna characterization experiment was performed at 1.3 GHz *by measuring the received power as a function of antenna separation and angular orientation. The received power decreased with increasing separation between the transmitting and receiving antennas, showing the expected distance dependence.*

The co-polarization measurements gave distinct E-plane and H-plane beam patterns, confirming the directional response of the antenna. A finite response was also observed in the cross-polarization measurements, indicating that complete polarization isolation was not achieved under the experimental conditions.

The E-plane half-power beamwidth was measured to be 111.3° *(*1.942 rad*). For the H-plane, only one clear* -3 dB *crossing was obtained, so the HPBW could not be determined directly from the measured data. Assuming approximate symmetry about the* 0° *direction gives an estimated H-plane HPBW of* 13.8°*. Using this value, the antenna directivity was estimated to be* $D \approx 26.9$*, corresponding to approximately* 14.3 dBi*.*

The VSWR experiment was also carried out to examine the impedance-matching behaviour at different frequencies. At 120 MHz*, a VSWR of* 11.25 *and a return loss of* 1.55 dB *were obtained, indicating poor impedance matching. At* 300 MHz*, the VSWR decreased to* 1.50 *and the return loss increased to* 13.98 dB*, indicating better matching at this frequency. These measurements show that the impedance-matching condition varies with frequency. However, the two measurement points are insufficient to determine the exact resonant frequency or the complete impedance bandwidth.*

Overall, the experiment demonstrated antenna characterization through received-power measurements, radiation-pattern analysis, polarization measurements and VSWR calculations. The main limitations included the 20° *angular sampling interval, possible alignment and measurement errors, and the mismatch between the* 0° *and* 360° *measurements in the H-plane. The H-plane beamwidth and corresponding directivity are therefore approximate estimates. The VSWR results are also limited to the two frequencies measured. More closely spaced angular and frequency measurements would help provide a more accurate characterization of the antenna.*

XI. REFERENCES

1. *Lab Mannual of SPA617*
2. *Wikipedia*
3. *AI tool for more conceptual understanding.*

XII. DECLARATION OF AI USAGE

I have used AI tools in the following ways for the completion of my lab report:

- *Formatting the theory content into various sub sections and concise writing of the same after I studied them from various sources. It helped me compile them well.*
- *Formatting of graph in python (lines along with labels) and the algorithm to find the cut off frequency from the graph.*

However ,the calculations, error analysis, data acquisition and processing, conclusions, key takeaways, results were all self written from scratch and there is absolutely no AI aid taken over those sections.

XIII. ACKNOWLEDGMENT

I would like to thank our course instructor Dr. Mugundhan V. , our REO - Jayasandhya Meenakshinathan and our TA - Banhiman Ghosh, without their help this lab experiment would have been very difficult to complete.